A Two-Isogeny Descent Attached to Campbell's Eight-Term Family

Codex (GPT-5.6-sol)

Revised 4 September 2026

Abstract

Starting from Campbell's cubic Weierstrass family with eight rational points whose x-coordinates are $0, \ldots, 7$, we reconstruct the quartic condition for the ninth candidate and record every rational specialization boundary. We identify its Jacobian and rational 2-torsion, then perform an exact finite local calculation for the dual 2-isogeny descents. The resulting Selmer groups have dimensions 3 and 2, respectively, which gives Mordell–Weil rank at most 3. An explicit integral change gives a global minimal semistable model of conductor 301245307115205810. We also compute the full $\mathbf{Q} \times K$ representative of the binary-quartic cubic invariant and its rational 2-torsion projection. A versioned supplement records the 512 local cells, common-parameter local witnesses, exact generators, certificates, and regression tests. Uniform valuation arguments show that the E'-side classes surviving both $\mathbf{Q}_2$ and $\mathbf{Q}_3$ are exactly the four Selmer classes; together with the analogous E-side two-place gate, this replaces the decisive portions of the local table by proofs. The same-parameter genus-5 fibre product is everywhere locally soluble. Our calculation is deliberately narrower than the original extension problem: it neither produces nor excludes a ninth rational point on Campbell's family, claims no full 2-Selmer group or Mordell–Weil rank equality, and does not evaluate a Cassels–Tate pairing. The results are therefore a reproducible finite descent computation and a rigorously delimited input to further arithmetic study of the ninth-value curve.

Keywords. elliptic curves; 2-isogeny descent; Selmer groups; arithmetic progressions; local solubility.

Mathematics Subject Classification (2020). Primary 11G05; Secondary 11G30, 11Y50.

1 Prior work and contribution boundary

Campbell's article [1] constructs the eight-term family used below. His dissertation [2, pp. 69–70] also records, for curves $y^2 = x^3 + ax^2 + bx$, the standard rational 2-isogeny, its dual, the quartics $z^2 = dU^4 + aU^2V^2 + (b/d)V^4$, the two isogeny Selmer groups, and the resulting rank formula. Thus neither rational 2-isogeny descent nor this covering formalism is claimed as new here. Our contribution is the exact finite computation for the particular index-8 ninth-value Jacobian: the two displayed isogeny Selmer groups and the consequent upper bound rank $E(\mathbf{Q}) \leq 3$, together with the stated certificates and arithmetic invariants. Campbell's rank-2 computation concerns his parameter curve D, not this Jacobian.

MathSciNet and zbMATH records and exact invariant searches did not reveal a prior computation of these particular groups. This is a coverage-limited not-found result, not a claim of priority: the full MathSciNet citation graph was unavailable, old zbMATH reference metadata are incomplete, and the target large composite conductor lies outside the published complete ranges of LMFDB and Cremona's ecdata tables.

2 The Campbell quartic

Campbell [1, Corollary 2.4 and Theorem 2.5] uses the parameter change

$$t = \frac{6m^2 - 126m - 285360}{m^2 - 72256}.$$

Its denominator never vanishes for $m \in \mathbf{Q}$, since $268^2 < 72256 < 269^2$. In the normalization used here put

$$g_m(x) = g_3x^3 + g_2x^2 + g_1x + g_0,$$

where

$$\begin{aligned}
g_3 &= -18816m^4 + 677376m^3 + 1922543616m^2 - 48944480256m - 40678301368320,\\
g_2 &= 236896m^4 - 9821952m^3 - 22598349824m^2 + 508953231360m + 520252184657920,\\
g_1 &= -958800m^4 + 40985280m^3 + 89932669440m^2 - 1957723729920m - 2113363439616000,\\
g_0 &= 1292769m^4 - 57304800m^3 - 118795148928m^2 + 2647001548800m + 2758336954896384.
\end{aligned}$$

Exact expansion gives $g_m(j) = q_j(m)^2$ for $0 \le j \le 6$; the square roots are listed in Table 1. The

Table 1: The first seven prescribed square roots in Campbell's family

j	$q_j(m)$
0	$3(379m^2 - 8400m - 17506624)$
1	$743m^2 - 17136m - 33534272$
2	$415m^2 - 11088m - 16946752$
3	$3(67m^2 - 3696m - 494656)$
4	$209m^2 - 17136m + 5050432$
5	$263m^2 - 25200m + 10631872$
6	$63(m^2 - 2352m + 72256)$

eighth condition is

$$\begin{aligned}
Y^2 = D(m) = &-264815m^4 - 19343520m^3 + 62846856064m^2\\
&- 2906312951808m - 495507443511296,
\end{aligned}$$

because $g_m(7) = D(m)$. The candidate immediately following the eight indices $0, \ldots, 7$ has index 8, and direct expansion gives $g_m(8) = H(m)$, where

$$\begin{aligned}
H(m) = &-850079m^4 - 11210976m^3 + 138714149248m^2\\
&- 5501355374592m - 1679721044504576
\end{aligned}$$

and put $C_H : z^2 = H(m)$. Thus Campbell's construction supplies eight prescribed squares precisely when (m, Y) lies on $Y^2 = D(m)$, and $z^2 = H(m)$ tests the ninth candidate at the same m.

There are no omitted rational specialization boundaries. The leading coefficient factors as

$$g_3 = -18816(m^2 - 72256)(m^2 - 36m - 29920),$$

and its two quadratic factors have no rational roots. After division by -65028096, the discriminant of g_m in x is a primitive degree-16 polynomial irreducible modulo 53, hence it has no rational zero. The point $m = \infty$ is not rational on $Y^2 = D(m)$, since its leading equation would be $Y^2 = -264815M^4$. A zero of D or H merely gives $Y = 0$ or $z = 0$ at one of the distinct x-coordinates; it does not make g_m singular. Finally, D and H are separable and coprime, so their smooth fibre product has genus 5.

Lemma 1 (same-parameter local solubility)**.** *The smooth curve*

$$Y^2 = D(m), \qquad z^2 = H(m)$$

has points over $\mathbb{R}$ *and every* $\mathbf{Q}_p$.

Proof. The supplement stores a common m and both square roots at $\mathbb{R}$, $\mathbf{Q}_2$, every odd prime below 101, and every odd prime dividing the two quartic discriminants or their resultant. The branch-bad set is

$$\{2, 3, 5, 7, 17, 19, 31, 59, 8599, 71699, 898543, 23037169, 339106321, 1153266911\}.$$

At every remaining prime $p \geq 101$ the fibre product has smooth genus-5 reduction, and the Weil bound $\#C(\mathbf{F}_p) \geq p+1-10\sqrt{p} > 0$ supplies a smooth point, which lifts by Hensel's lemma. The exact witness table is the object `same_m_local_certificates` in the manifest-bound certificate. □

In particular C_H is everywhere locally soluble. Nothing here asserts that the fibre product or C_H has a rational point.

For a binary quartic

$$g(X, Z) = a_0X^4 + a_1X^3Z + a_2X^2Z^2 + a_3XZ^3 + a_4Z^4$$

we use

$$I = 12a_0a_4 - 3a_1a_3 + a_2^2, \quad J = 72a_0a_2a_4 - 27a_0a_3^2 - 27a_1^2a_4 + 9a_1a_2a_3 - 2a_2^3.$$

The resulting Jacobian is $y^2 = x^3 - 27Ix - 27J$; see [4, Lemma 2.1] for the binary-quartic description of the full 2-Selmer group.

3 The cubic algebra and the Campbell class

Let $L = \mathbf{Q}[\varphi]/(\varphi^3 - 3I\varphi + J)$. For the above normalization, the cubic invariant of g is

$$z(g) = \frac{4a_0\varphi + 3a_1^2 - 8a_0a_2}{3} \in L^\times/L^{\times 2}; \tag{1}$$

this is the invariant used in [4, Section 2]. In the present case

$$L \simeq \mathbf{Q} \times K, \qquad K = \mathbf{Q}(w), \quad w^2 = D, \quad D = 1434501462453361 = 59 \cdot 71699 \cdot 339106321.$$

The quadratic resolvent factor is

$$T^2 + 269378023424T - 3600948712181056353075 2,$$

whose discriminant is 12288^2D. Thus its roots are $-134689011712 \pm 6144w$. Although $\mathcal{O}_K = \mathbb{Z}[(1+w)/2]$, all displayed coordinates below use the $\mathbf{Q}$-basis $(1, w)$.

Proposition 2. *The two components of the cubic invariant of H are*

$$z_{\mathbf{Q}} = 9250179026780160 = 35 \cdot 16257024^2$$

and

$$z_K = 467235380575281152 - 6963847168w = 64^2(114071137835762 - 1700158w).$$

Moreover

$$N_{K/\mathbf{Q}}(114071137835762 - 1700158w) = 35 \cdot 15915620907648^2,$$

and the full norm is

$$z_{\mathbf{Q}}N_{K/\mathbf{Q}}(z_K) = 370930568702719434109747 20^2.$$

Proof. Insert the five coefficients of H into (1), then evaluate in the rational and quadratic factors of L. Squaring the three displayed integers gives the claimed identities. These are integer identities, independently recomputed by the accompanying regression test. $\square$

The rational factor in $L = \mathbf{Q} \times K$ is the cohomological factor induced by the rational 2-torsion point. On scaling

$$x_{\text{big}} = 64^2 x, \qquad y_{\text{big}} = 64^3 y$$

and translating $X = x + 197298357$, the Jacobian becomes

$$E : y^2 = X^3 - 591895071X^2 + 58536289153843200X. \tag{2}$$

Since $-3\varphi_0 = 64^2(-197298357)$ for the rational resolvent root $\varphi_0 = 269378023424$, the rational projection of $[H] \in H^1(\mathbf{Q}, E[2])$ is the X-Kummer squareclass [35]. This assertion concerns one projection only: C_H is not thereby identified with the even quartic C_{35} below.

3.1 A global minimal model and conductor

The model (2) is convenient for the rational 2-isogeny but is not globally minimal. The admissible change

$$X = 36x, \qquad y = 216y_0 + 108x \tag{3}$$

gives

$$E_{\min} : \quad y_0^2 + xy_0 = x^3 - 16441530x^2 + 45166889779200x. \tag{4}$$

Equivalently, the admissible parameters are $(u, r, s, t) = (6, 0, 3, 0)$, and the a-invariants of (4) are $(1, -16441530, 0, 45166889779200, 0)$.

Proposition 3. *Equation* (4) *is a global minimal model. Its invariants are*

$$c_4 = 2157171698920561, \qquad c_6 = 70577985500751764077559,$$

and

$$\begin{aligned}\Delta_{\min} &= 29264517423971780756539747446869616230400000\\ &= 2^{28}3^{16}5^4 7^{10} \cdot 59 \cdot 71699 \cdot 339106321.\end{aligned}$$

The curve is semistable, with multiplicative reduction of type $I_{v_p(\Delta_{\min})}$ at each prime in

$$\{2, 3, 5, 7, 59, 71699, 339106321\},$$

and good reduction elsewhere. The multiplicative reduction at 2 and at 3 is split. In particular,

$$N_E = 2 \cdot 3 \cdot 5 \cdot 7 \cdot 59 \cdot 71699 \cdot 339106321 = 301245307115205810.$$

Finally,

$$j(E) = \frac{10038160648206649953061393462836377818780518481}{29264517423971780756539747446869616230400000}.$$

Proof. Substitution in the general admissible-change formulas gives (4). Direct calculation gives the displayed c_4, c_6 and discriminant and verifies $c_4^3 - c_6^2 = 1728\Delta_{\min}$; the original invariants are respectively $6^4, 6^6, 6^{12}$ times these invariants. Moreover

$$\gcd(c_4, \Delta_{\min}) = 1.$$

At every prime dividing $\Delta_{\min}$ we therefore have $v_p(c_4) = 0$. An integral Weierstrass equation that were nonminimal at p would have $p^4 \mid c_4$, so this proves p-minimality. This argument is valid without change at $p = 2$ and $p = 3$.

For an explicit check at these two residue characteristics, move the right side of (4) to the left. Both 16441530 and 45166889779200 vanish modulo 2 and modulo 3, so the reduced affine equation is

$$F(x, y) = y^2 + xy - x^3 = 0.$$

Modulo 2 its partial derivatives are $(F_x, F_y) = (y + x^2, x)$, and modulo 3 they are $(F_x, F_y) = (y, 2y + x)$. Thus in either case the unique affine singular point is $(0, 0)$; the point at infinity is smooth because the Z-derivative of the homogenized equation is nonzero there. At $(0, 0)$ the quadratic tangent cone is

$$y^2 + xy = y(y + x),$$

whose two linear factors are distinct over both $\mathbf{F}_2$ and $\mathbf{F}_3$. Hence the multiplicative reduction at 2 and 3 is split. The criterion $v_p(\Delta) > 0$, $v_p(c_4) = 0$ gives multiplicative reduction at every other bad prime as well, with conductor exponent one. At every other prime the discriminant is a unit. The conductor and Kodaira symbols follow, and $j = c_4^3/\Delta_{\min}$. See [6, Chapter VII, Section 5] for the reduction criteria. □

This calculation supplies exact isomorphism-class search keys. It is not a rank computation, and the absence of a database row at this large composite conductor is not evidence for novelty.

4 The rational two-isogeny

Write $A = -591895071$ and $B = 58536289153843200$. The quotient by $(0, 0)$ gives

$$\phi : E \longrightarrow E', \qquad E' : y^2 = X^3 - 2AX^2 + (A^2 - 4B)X,$$

so explicitly

$$E' : y^2 = X^3 + 1183790142X^2 + 116194618458722241X. \tag{5}$$

We denote the dual isogeny by $\widehat{\phi} : E' \to E$. The standard isogeny descent and its covering quartics are described, for example, in [6, Chapter X, Section 4] and [3, Chapter 14]; higher descent with rational 2-torsion is treated directly in [5].

Lemma 4 (support)**.** *For $E_{a,b} : y^2 = x^3 + ax^2 + bx$, every class in the x-Kummer isogeny Selmer group has a squarefree representative supported on -1 and the prime divisors of b. The class d is represented by*

$$C_d : \quad N^2 = dU^4 + aU^2V^2 + (b/d)V^4. \tag{6}$$

Proof. Let $p \nmid b$ divide a squarefree representative d, and after weighted projective scaling take $U, V \in \mathbb{Z}_p$ primitive. In the original equation (6), if V were a unit, then $(b/d)V^4$ would be the unique term of valuation -1 on the right, which cannot be the valuation of the square N^2. Hence $p \mid V$, so primitivity makes U a unit. The right side of (6) then has valuation exactly 1, again impossible for a square. Notice that this argument does not assume in advance that N is p-integral. Thus $v_p(d)$ is even outside b, and squareclass reduction gives the stated support. □

Proposition 5 (two-place E-side gate)**.** *For the curve E in (2), let d run through the 32 signed squarefree classes supported on $\{2, 3, 5, 7\}$. Then*

$$C_d(\mathbf{Q}_{59}) \neq \varnothing \iff C_d(\mathbf{Q}_{71699}) \neq \varnothing.$$

These equivalent conditions hold exactly for

$$d \in \left\{ \begin{matrix} -210, -70, -42, -30, -14, -10, -6, -2, \\ 1, 3, 5, 7, 15, 21, 35, 105 \end{matrix} \right\}.$$

Moreover $C_d(\mathbf{R}) \neq \varnothing$ exactly when $d > 0$. Consequently the classes that survive the real place together with either one of these two finite places are exactly

$$\{1, 3, 5, 7, 15, 21, 35, 105\}.$$

Proof. Write $a = -591895071$, $b = 58536289153843200$, and

$$\delta = a^2 - 4b = 3^4 \cdot 59 \cdot 71699 \cdot 339106321.$$

For $p = 59$ or 71699 one has $v_p(\delta) = 1$, while every listed d and $2b$ are p-adic units. The covering quartic satisfies

$$4dF_d = (2dU^2 + aV^2)^2 - \delta V^4, \qquad F_d = dU^4 + aU^2V^2 + (b/d)V^4. \tag{7}$$

If $(d/p) = 1$, the choice $(U : V) = (1 : 0)$ gives a point because d is a square in $\mathbf{Q}_p$. Conversely suppose $(d/p) = -1$ and normalize $U, V \in \mathbf{Z}_p$ to be primitive. If $p \mid V$, then $F_d \equiv dU^4 \pmod{p}$ is a nonsquare unit. If V is a unit and $2dU^2 + aV^2$ is a unit, (7) shows that F_d again has squareclass d modulo p. In the remaining case, $p \mid 2dU^2 + aV^2$; the right side of (7) then has valuation exactly one, so $v_p(F_d) = 1$. Each case contradicts $F_d = N^2$.

Quadratic reciprocity gives, at both primes,

$$\left(\frac{-1}{p}\right) = \left(\frac{2}{p}\right) = -1, \qquad \left(\frac{3}{p}\right) = \left(\frac{5}{p}\right) = \left(\frac{7}{p}\right) = 1.$$

This yields the displayed 16-class local list. Finally, if $d > 0$ then $(U : V) = (1 : 0)$ gives a real point. If $d < 0$, all three coefficients $d, a, b/d$ of F_d are negative, so $F_d < 0$ for every nonzero real pair (U, V). Intersecting the two classifications gives the eight positive odd divisors of 105. □

Proposition 6 (two-place E'-side gate)**.** *Put*

$$A' = 1183790142, \qquad B' = 1161946184587222241 = 3^4 D, \qquad q = 339106321.$$

For the 32 signed squarefree classes d supported on $\{3, 59, 71699, q\}$, write

$$C'_d : \quad N^2 = dU^4 + A'U^2V^2 + (B'/d)V^4.$$

Then

$$C'_d(\mathbf{Q}_2) \neq \varnothing \iff d \equiv 1 \pmod{8},$$

and its eight solutions among the supported classes are

$$\{1, q\}\{1, 3 \cdot 59, 3 \cdot 71699, 59 \cdot 71699\}.$$

Moreover

$$C'_d(\mathbf{Q}_3) \neq \varnothing \iff v_3(d) = 0 \text{ and } d \equiv 1 \pmod{3},$$

and the corresponding eight classes are

$$\{1, q\}\{1, 59 \cdot 71699, -59, -71699\}.$$

Consequently simultaneous solubility at 2 and 3 leaves exactly

$$\{1, 4230241, 339106321, D\}.$$

No single finite place in the stored eight-place matrix leaves only these four classes; thus two is the minimum number of finite places that recovers this four-class set from that matrix. The only other minimal pair is $\{2, 5\}$.

Proof. Write $A' = 2c$, so $c = 591895071 \equiv 7 \pmod 8$. Direct integer factorization gives

$$B' = 3^4 \cdot 59 \cdot 71699 \cdot q \equiv 1 \pmod 8, \qquad (A')^2 - 4B' = 2^{22}k,$$

where $k = 223298222175$ is odd. If F'_d denotes the quartic on the right of the equation for C'_d, coefficient comparison gives

$$4dF'_d = (2dU^2 + A'V^2)^2 - ((A')^2 - 4B')V^4.$$

Thus every point satisfies

$$dN^2 = T^2 - 2^{20}kV^4, \qquad T = dU^2 + cV^2. \tag{8}$$

Normalize a putative 2-adic point so that $U, V \in \mathbf{Z}_2$ are primitive. Suppose $d \not\equiv 1 \pmod 8$. When U, V have opposite parity, T is odd. When both are odd, their squares are 1 modulo 8, and $T \equiv d + 7 \pmod 8$; hence

$$v_2(T) = 1, 2, 1 \quad \text{as} \quad d \equiv 3, 5, 7 \pmod 8.$$

In every case $v_2(T) \leq 2$, and therefore

$$v_2\left(2^{20}kV^4/T^2\right) \geq 16.$$

Every element of $1 + 8\mathbf{Z}_2$ is a square (equivalently, apply the strengthened Hensel criterion at the approximate square root 1). Equation (8) consequently makes dN^2 a nonzero square in $\mathbf{Q}_2$, forcing d itself to be a square, a contradiction. Conversely, if $d \equiv 1 \pmod 8$, then d is a square in $\mathbf{Q}_2$: for $f(X) = X^2 - d$, one has $v_2(f(1)) \geq 3 > 2v_2(f'(1)) = 2$. The point $(U : V : N) = (1 : 0 : \sqrt{d})$ proves sufficiency.

For the 3-adic assertion, note that $v_3(A') = 2$, $A'/9 \equiv 1 \pmod 3$, and $B'/3^4 = D \equiv 1 \pmod 3$. Again normalize $U, V \in \mathbf{Z}_3$ to be primitive. If $v_3(d) = 1$, then the three terms of F'_d show that $v_3(F'_d) = 1$ when U is a unit. If $3 \mid U$, then V is a unit and the last term is the unique term of least valuation, giving $v_3(F'_d) = 3$. Neither valuation can occur for a square.

Now let $v_3(d) = 0$. If U is a unit, reduction modulo 3 gives $F'_d \equiv dU^4$, so solubility requires $d \equiv 1 \pmod 3$. If $U = 3^r u$ with $r \geq 1$ and u, V units, divide by 3^4. For $r \geq 2$ the result is congruent to $(D/d)V^4 \equiv d^{-1} \pmod 3$. For $r = 1$ it is congruent to

$$du^4 + (A'/9)u^2V^2 + (D/d)V^4 \equiv d + 1 + d^{-1} \pmod 3.$$

When $d \equiv -1 \pmod 3$, both displayed units are -1, so no square is possible. Conversely $d \equiv 1 \pmod 3$ is a square in $\mathbf{Q}_3$ by ordinary Hensel lifting of the root 1 of $X^2 - d$ modulo 3, and the same point $(1 : 0 : \sqrt{d})$ works.

Finally $59 \equiv 71699 \equiv 3 \pmod 8$, $q \equiv 1 \pmod 8$, while $59 \equiv 71699 \equiv -1 \pmod 3$ and $q \equiv 1 \pmod 3$. These residues give the two factored eight-class lists and their four-class intersection. The last minimality statement is an exact comparison with all seven finite place columns of the 512-cell certificate: the smallest one-place survivor count is eight, and precisely the pairs $\{2, 3\}$ and $\{2, 5\}$ have the displayed four-class intersection. This comparison is independently recomputed by the standard-library regression script; the local classification itself is proved above without using the table. $\square$

Lemma 7 (good primes)**.** *Fix a supported squarefree d. If a finite prime p does not divide $2b(a^2 - 4b)$, then the smooth projective model of C_d has a $\mathbf{Q}_p$-point.*

Proof. The binary-quartic discriminant of the right side of (6) is

$$16b(a^2 - 4b)^2.$$

Thus C_d has smooth geometrically connected genus-one reduction at p. In this application such a p is at least 5, so the Hasse bound gives $\#C_d(\mathbf{F}_p) \geq p + 1 - 2\sqrt{p} > 0$. Every point of the smooth reduction lifts by Hensel's lemma. $\square$

5 Exact local classification

The only places required on either side are

$$S = \{\infty, 2, 3, 5, 7, 59, 71699, 339106321\}.$$

Lemma 4 gives 32 signed candidates on each side, and Lemma 7 proves that no unlisted place can remove one. The certificate contains all $64 \times 8 = 512$ local cells. Propositions 5 and 6 now give uniform proofs for the decisive E- and E'-side reductions. A YES cell contains an exact squareclass witness; a NO cell is either an exhaustive weighted-projective computation modulo the displayed prime power or the valuation-normalized double-root obstruction. All 56 cells that had previously been unresolved are now split into 24 YES cells at 3 and 32 NO cells at 59 or 71699.

Table 2: Exact local classification of the two isogeny-descent candidate sets

side	candidates	first real NO	first finite NO	survivors	survivor set
E	32	16	8 at 59	8	$1, 3, 5, 7, 15, 21, 35, 105$
E'	32	0	24 at 2, 4 at 3	4	$1, 4230241, 339106321, D$

The complete matrix has 384 YES and 128 NO cells. Its JSON SHA-256 is

`fff2f35f398c2d14227d9b032d205e24d5332a39346487c76180cdf805ba32c9.`

The structured $\mathbf{Q} \times K$ and Selmer certificate has SHA-256

`7fbd4d00435d2f3ef7bd74b0f87822b8e6fb3b893e2023ce9b5741c411538e18.`

This is the clean Round-04 certificate: it contains only the exact Selmer, $\mathbf{Q} \times K$, scaling, and proved Mordell–Weil-image data. The superseded opposite-side pairing proposal is absent; its failure is recorded only in the separate Round-05 negative audit. Both files, the same-m certificate, their generators and regression tests are located by supplement release

`paper-elliptic-campbell-supplement-v0.6.1`

with root manifest

`PAPER_ELLIPTIC_SUPPLEMENT_MANIFEST.json`.

The manifest SHA-256 is

`583ec02b13c96824df5563d0dc7cc0312b7fbb33e5b7860b4f692b1fbed228fd.`

The manifest records byte lengths, roles, evidence eligibility, Python and SymPy versions, and exact reproduction commands. The current working tree is public at https://github.com/lixiang90/vibemath; its immutable archive identifier remains to be supplied.

Theorem 8. *With E-side classes represented by the X-Kummer map on E and E'-side classes by that on E', one has*

$$\mathrm{Sel}^{\widehat{\phi}}(E'/\mathbf{Q}) = \langle 3, 5, 7\rangle, \qquad \mathrm{Sel}^{\phi}(E/\mathbf{Q}) = \langle 4230241, 339106321\rangle.$$

Their $\mathbf{F}_2$-dimensions are respectively 3 *and* 2*.*

Proof. The two support lists are exhaustive by Lemma 4; all good places are automatic by Lemma 7. Proposition 5 removes every E-side class outside the first displayed group, and Proposition 6 does the same on the E' side. The exact witnesses in the certificate verify that every displayed class is soluble at the remaining bad places, yielding exactly the survivor sets in Table 2. Direct squareclass multiplication verifies that the two sets are the groups generated as displayed. □

The exact sequence underlying descent by a degree-2 isogeny gives

$$2^{\operatorname{rank} E(\mathbf{Q})} = \frac{|E'(\mathbf{Q})/\phi E(\mathbf{Q})|\,|E(\mathbf{Q})/\widehat{\phi} E'(\mathbf{Q})|}{|E(\mathbf{Q})[\phi]|\,|E'(\mathbf{Q})[\widehat{\phi}]|};$$

see [7, pp. 368–369]. Both denominator groups here have order 2, and the two Mordell–Weil quotients inject into the two Selmer groups.

Corollary 9. $\operatorname{rank} E(\mathbf{Q}) \leq 3$.

6 Correction of the proposed pairing gate

For clarity we record the relevant cohomological direction. The standard pairing that upgrades a descent by $\widehat{\phi}$ to a full 2-descent is an alternating pairing

$$\operatorname{Sel}^{\widehat{\phi}}(E'/\mathbf{Q}) \times \operatorname{Sel}^{\widehat{\phi}}(E'/\mathbf{Q}) \longrightarrow \mathbf{Q}/\mathbb{Z}, \tag{9}$$

whose radical is the image of the appropriate full 2-Selmer group; see [7, Equation (2)]. In particular, it is a pairing on one isogeny Selmer group squared, not a pairing between the two groups in Theorem 8. The full Cassels–Tate pairing on binary quartics likewise has domain $\operatorname{Sel}^{(2)}(E/\mathbf{Q}) \times \operatorname{Sel}^{(2)}(E/\mathbf{Q})$ [4, Theorem 3.1].

Proposition 10 (pairing correction)**.** *The expression formerly denoted $\langle 35, 4230241\rangle_{CT}$ in this project is not defined by either of the standard pairings just cited: 35 and 4230241 are coordinates in opposite isogeny Selmer groups. Consequently its proposed tangent-line formula cannot be used to deduce $C_H(\mathbf{Q}) = \varnothing$.*

Proof. The domain mismatch follows directly from (9) and Theorem 8. There is a second consistency check. By Lemma 1, C_H is everywhere locally soluble, so it represents a class of $\operatorname{Sel}^{(2)}(E/\mathbf{Q})$, and its projection is 35. Hence 35 already lies in the image whose radical is tested by (9); any correctly defined same-side lifting pairing must annihilate it. ☐

The arithmetic that exposed the problem is still useful as a negative certificate. For the standard even quartic

$$N^2 = 35U^4 - 591895071U^2V^2 + 1672465404395520V^4$$

the associated conic in $R = U^2, S = V^2$ has the rational point

$$(R_0, S_0, N_0) = (16257024, 1, 36058176).$$

A primitive tangent there is

$$L_{35} = 60677401R - 697502396215296S - 8012928N.$$

Both identities follow by direct substitution. If one nevertheless evaluates the rejected expression $(L_{35}(P_v), 4230241)_v$, then the two Hensel branches give the exact residues in Table 3.

Each listed N squares to the quartic right side modulo the listed modulus. Thus independent local branch choices yield both possible products. This does not compute a Cassels–Tate value; it proves that the formerly proposed bare expression lacks the cochain or denominator data needed for well-definedness.

Table 3: Branch dependence of the rejected local Hilbert-symbol expression

p	modulus	N	L_{35}	$v_p(L_{35})$	Hilbert symbol
59	59^3	56226	127303	0	-1
59	59^3	149153	145730	1	$+1$
71699	71699^3	152989401412805	252464922339130	0	-1
71699	71699^3	215596989132294	195462237955773	1	$+1$

7 The exact remaining full-descent problem

To test the Campbell class itself one must now do one of the following equivalent full-descent computations.

1. Compute a basis of $\mathrm{Sel}^{(2)}(E/\mathbf{Q})$ as everywhere locally soluble binary quartics g_i with the same I, J; identify $[H]$ by the full $z(H) \in (\mathbf{Q} \times K)^\times/(\mathbf{Q} \times K)^{\times 2}$; and compute $\langle [H], [g_i] \rangle_{CT}$ by the full binary-quartic formula of [4, Theorem 3.1].

2. Compute the locally soluble 4-coverings lying above C_H. An empty output proves that its 2-Selmer class does not lift and hence that $C_H(\mathbf{Q}) = \varnothing$; a nonempty output does not by itself produce a rational point. This is the precise role of a four-descent [8, Section "Four-Descent"].

The accompanying Magma input freezes both routes, but the manifest marks it `UNEXECUTED_FROZEN_CANDIDATE_INPUT` and `mathematical_evidence_eligible=false`. No transcript or trusted binary hash is present. Therefore no full 2-Selmer dimension, Cassels–Tate value, Mordell–Weil rank equality, or rational-point conclusion is claimed.

8 Finite theorem and reproducibility boundary

Theorem 11 (finite Campbell–Jacobian calculation)**.** *For the binary quartic H obtained at index* 8 *from Campbell's Theorem 2.5, the two isogeny Selmer groups are those in Theorem 8; hence* $\operatorname{rank} E(\mathbf{Q}) \leq 3$. *Its cubic-algebra invariant is the pair of Proposition 2, with rational* 2*-torsion projection* [35]. *Its global minimal model, discriminant, local reduction types, and conductor are those of Proposition 3. Moreover the same-parameter genus-*5 *fibre product is everywhere locally soluble.*

This is a finite, certificate-backed theorem. Exact-coefficient searches did not locate a published computation of these particular Selmer groups; that is a not-found report, not proof of novelty or priority. The theorem neither asserts the existence or nonexistence of a rational ninth point nor computes the full 2-Selmer group or any Cassels–Tate value. No independent second elliptic-curve CAS was available; the exact standard-library local proof and compatibility test are not represented as an independent external reproduction. In particular, the local theorem does not decide the ninth-point problem. A fully specified Sage `test_els` reproduction plan is included in the research archive, but Sage, Magma, and PARI/GP were not run and no second-CAS result is claimed.

Statements and Declarations

Funding. No external funding was received for this work.

Competing interests. The author declares no competing interests.

Author contributions. Codex (GPT-5.6-sol): conceptualization, formal analysis, methodology, software, validation, reproducibility, and writing of the original and revised manuscript.

Ethics approval, consent to participate, and consent for publication. Not applicable: this article contains only mathematical research and uses no human participants, personal data, or animals.

Data and code availability. The exact generators, JSON certificates, and regression tests are identified by supplement release `paper-elliptic-campbell-supplement-v0.6.1`. The current working version is public at https://github.com/lixiang90/vibemath. The repository-root manifest binds each completed research round. No DOI or immutable preservation identifier is asserted because no actual submission is requested.

Computational authorship and disclosure. The sole named author is Codex (GPT-5.6-sol), a computational research agent. It performed the symbolic-algebra scripting, exact-arithmetic checks, test drafting, literature-query formulation, and prose preparation. The present artifact is intended to reach research and referee readiness; no claim is made that an AI system satisfies any particular journal's legal or editorial authorship policy, and this work has not been submitted for publication.

Supplementary information. Online Resource 1: deterministic verification archive containing the exact source, frozen reference PDF, generators, certificates, data dictionary, and regression tests identified by release root `paper-elliptic-campbell-local-release-v0.6.3`. The unexecuted Magma input is included solely as an ineligible candidate input and is not evidence for any theorem. No affiliation, postal address, email, ORCID, or immutable archive identifier is asserted.

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